FedShield-IDS: A Federated, Adversarially-Robust Intrusion Detection System for Heterogeneous IoT Networks

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***Abstract*—The rapid expansion of Internet of Things (IoT) devices has created an expansive, heterogeneous attack surface across critical infrastructure. While deep learning-based Network Intrusion Detection Systems (NIDS) exhibit high precision in closed settings, they face two systemic vulnerabilities when deployed at the edge: (i) strict data localization requirements that prohibit centralized aggregation of raw packet telemetry; and (ii) dual-vector adversarial fragility, wherein subtle flow perturbations bypass detection at inference time and compromised edge clients poison global parameter updates during training. Crucially, conventional Byzantine-resilient aggregation rules fail under statistical heterogeneity (non-IID data), routinely conflating legitimate domain-specific traffic with malicious parameter poisoning.**

In this paper, we propose FedShield-IDS, a federated intrusion detection framework that couples client-side Projected Gradient Descent (PGD) adversarial training with server-side kernel-normalized anomaly-aware aggregation. On edge clients, local convolutional neural networks are regularized against bounded ℓ_∞ feature perturbations constrained by physical network flow invariants. At the central coordinator, incoming parameter updates undergo layer-wise Frobenius normalization and cosine directional consensus against coordinate-wise median trajectories, systematically isolating Byzantine manipulators without penalizing non-IID benign clients. Evaluations on real network flow telemetry from the benchmark CIC-IDS2017 dataset demonstrate that FedShield-IDS achieves 94.48% clean classification accuracy under Dirichlet heterogeneity ($\beta = 0.5$), suppresses black-box evasion Attack Success Rates (ASR) from 34.64% to 5.60% (a 29.04 percentage point defensive margin), and neutralizes 100% of Byzantine model-poisoning updates while sustaining sub-0.12 ms per-flow edge inference latency.

***Index Terms*—Intrusion Detection System, Federated Learning, Adversarial Robustness, IoT Security, Byzantine Resilience, Deep Learning**

I. Introduction

The operational reality of edge Internet of Things (IoT) networks presents a stark security asymmetry: resource-constrained edge appliances running diverse, infrequently patched firmware are continuously exposed to automated, multi-vector cyber threats. Threat actors increasingly exploit the vulnerable perimeter through distributed denial-of-service (DDoS) botnets, lateral reconnaissance, and zero-day endpoint exploits. To defend these perimeters, deep learning (DL) has emerged as the premier paradigm for Network Intrusion Detection Systems (NIDS), with 1D-Convolutional Neural Networks (1D-CNNs) [1] and hybrid CNN-LSTM architectures [2] achieving benchmark detection accuracies exceeding 98%.

However, transitioning deep NIDS from centralized offline testbeds to operational IoT infrastructures exposes two fundamental, structurally coupled challenges:

1) Telemetry Privacy and Distributed Governance: Classical NIDS architectures mandate continuous exfiltration of raw packet captures or statistical flow metadata to a centralized Security Operations Center (SOC). In multi-tenant, municipal, and healthcare IoT environments, aggregating raw network flows violates strict data sovereignty mandates (e.g., EU GDPR Articles 25 and 32, HIPAA Security Rule, and NIS2 Directive) while exposing sensitive proprietary network topologies. Federated Learning (FL) [3], [4] resolves this privacy bottleneck by distributing model training: edge nodes compute local gradient updates on private flow partitions and transmit only parameter deltas to a central coordinator.

2) Dual-Vector Adversarial Vulnerability: While FL preserves data confidentiality, decentralizing optimization eliminates the perimeter trust boundary, exposing the learning pipeline to dual-surface adversarial manipulation:

- *Inference-Time Evasion:* Deep classifiers exhibit extreme sensitivity to adversarial input perturbations [5]. Adversaries utilizing decision-based or query-efficient black-box techniques [6] craft subtle flow perturbations (e.g., slight adjustments to packet length distributions or inter-arrival times) that evade detection without violating network protocol syntax.
- *Training-Time Byzantine Model Poisoning:* Decentralized parameter synthesis exposes the aggregation server to Byzantine manipulation [7], [8]. Malicious or hijacked IoT nodes can submit poisoned weight deltas ($\tilde{\Delta}_m$) designed to overwrite decision boundaries, establishing persistent backdoors for specific attack categories or destabilizing global model convergence.

The Non-IID Defense Dilemma: Existing literature investigates these vulnerabilities in isolation. Federated IDS frameworks typically assume benign, homogeneous client behavior

[4], whereas adversarial defense studies assume centralized, white-box environments [5]. Naively combining existing defenses exposes an acute dilemma: in realistic IoT ecosystems, statistical traffic distributions across heterogeneous device classes (e.g., high-throughput IP cameras versus intermittent environmental sensors) are non-IID (non-identically and independently distributed). Standard Byzantine-resilient aggregation rules (e.g., coordinate-wise median, Trimmed Mean, or Krum) rely on geometric distance clustering; under severe statistical heterogeneity, these heuristics routinely mistake legitimate, specialized client telemetry for adversarial poisoning. Conversely, uncoordinated client-side adversarial training amplifies gradient dispersion, causing robust updates from honest edge devices to be prematurely rejected by conventional server-side filters.

To resolve this impasse, an operational IoT NIDS must simultaneously guarantee: (i) strict raw-data localization, (ii) edge resilience against evasion perturbations, and (iii) server-side Byzantine resilience that distinguishes malicious manipulation from non-IID statistical heterogeneity.

A. Contributions

In this paper, we propose **FedShield-IDS**, a cohesive federated intrusion detection framework engineered for heterogeneous IoT ecosystems. FedShield-IDS harmonizes edge-level adversarial regularization with coordinator-level normalized directional filtering. Our core contributions are:

- **Integrated Defense-in-Depth Architecture:** We design an end-to-end framework combining client-side Projected Gradient Descent (PGD) adversarial training on compact 1D-CNNs (186K parameters) with a two-tier server-side aggregation protocol combining layer-wise Frobenius normalization and cosine directional consensus.
- **Heterogeneity-Preserving Byzantine Resilience:** We formulate a kernel-normalized anomaly scoring metric that scales parameter updates by layer-wise dispersion relative to the median update trajectory, preventing benign non-IID clients from being falsely rejected while systematically isolating Byzantine corruptions.
- **Formal Dual-Threat Characterization:** We formalize a comprehensive threat model capturing simultaneous inference evasion under strict ℓ_∞-bounded black-box query budgets and up to 20% Byzantine poisoning participants executing targeted model subversion across non-IID Dirichlet traffic partitions ($\beta = 0.5$).
- **Rigorous Empirical Validation on Real Hardware:** We benchmark FedShield-IDS on genuine flow traces from CIC-IDS2017 executed on edge testbeds. FedShield-IDS attains 94.48% clean accuracy, suppresses black-box evasion Attack Success Rate (ASR) from 34.64% down to 5.60% (a 29.04 percentage point defensive margin), blocks 100% of Byzantine poisoned updates across communication rounds, and sustains line-rate flow inspection with an inference latency below 0.12 ms per flow.

B. Paper Organization

The remainder of this paper is organized as follows: Section II synthesizes related work. Section III formalizes the system model and dual-threat formulation. Section IV details the FedShield-IDS architecture, including Algorithms 1 and 2. Section V presents empirical evaluation results, baseline comparisons, and ablation studies. Section VI discusses operational trade-offs, and Section VII concludes the paper.

II. Related Work

A. Deep Learning for Network Intrusion Detection

Machine learning-based network intrusion detection has historically grappled with operational deployment barriers. In their seminal work, Sommer and Paxson [9] established the canonical operational dilemmas of NIDS—specifically, the extreme asymmetry of false positive costs in production Security Operations Centers (SOCs) and the semantic gap between statistical anomalies and actionable exploits. Over the past decade, the rapid expansion of network throughput has shifted the paradigm from shallow anomaly detectors to deep learning architectures.

Recently, Hnamte and Hussein [10] demonstrated that compact one-dimensional convolutional neural networks (1D-CNNs) achieve superior spatial feature extraction over high-dimensional flow statistics while minimizing computational overhead for resource-constrained IoT gateways. Kodys et al. [1] corroborated the efficacy of convolutional topologies on IoT traffic, whereas Hossan et al. [2] deployed hybrid CNN-LSTMs to capture temporal packet inter-arrival dynamics. Addressing non-stationary traffic distributions, Nakıp and Gelenbe [11] formulated online self-supervised learners, and Wilkie et al. [12] leveraged metric contrastive learning for zero-day threat clustering. Advanced representation models such as CST-AFNet [13] incorporate dual-attention mechanisms, while AutoGraphAD [14] and Kale et al. [15] leverage graph autoencoders for unsupervised reconstruction. Nevertheless, these centralized frameworks mandate the aggregation of raw network telemetry at a central server and remain intrinsically vulnerable to inference-time evasion perturbations.

B. Federated Learning in Edge Security

Federated Learning (FL), pioneered by McMahan et al. [3] via Federated Averaging (FedAvg), enables collaborative model training across decentralized clients without centralizing raw telemetry, thereby satisfying stringent privacy mandates. However, decentralized optimization introduces vulnerability to Byzantine manipulation. Classical robust aggregation rules—including Krum [16], Trimmed Mean, and Coordinate-wise Median [17]—were formulated under the assumption of independent and identically distributed (IID) client telemetry. In realistic IoT ecosystems, statistical traffic distributions across heterogeneous device classes (e.g., smart meters versus surveillance cameras) are heavily non-IID [4]. Under severe statistical heterogeneity, geometric distance-based filters routinely mistake legitimate, specialized client updates for Byzantine attacks.

Furthermore, Bagdasaryan et al. [18] demonstrated that malicious participants can exploit parameter scaling to inject backdoors into FedAvg. Recent proposals, such as the dynamic neural model for 5G/6G slices by Neha and Bhatia [19], integrate adversarial learning at the edge but rely on vanilla federated averaging, leaving the global model defenseless against training-time Byzantine poisoning. Additionally, Shuvo and Hossain [20] proved that unencrypted edge telemetry allows adversaries to reconstruct model parameters, reinforcing the critical requirement for robust, localized parameter defense.

C. Adversarial Attacks and Defenses

Adversarial attacks, formalized by Goodfellow et al. [21] and Carlini and Wagner [22], demonstrate that deep neural networks are vulnerable to imperceptible gradient-guided input perturbations. In network telemetry, Chen and Liu [5] systematized adversarial evasion against deep flow classifiers, while Vo et al. [6] demonstrated that decision-based black-box attacks reliably evade flow-based classifiers with minimal feature manipulation. Unlike computer vision where perturbations span unconstrained continuous domains, adversarial network traffic morphing is constrained by protocol syntactics and physical transport feasibility.

To counter evasion, Madry et al. [23] formulated adversarial defense as a minimax optimization problem solved via Projected Gradient Descent (PGD). On the training-time front, data preprocessing tampering [24] and compression backdoors [8] threaten collaborative integrity. For server-side defense, Nasirigerdeh et al. [25] established kernel-normalized bounds to constrain outlier updates. FedShield-IDS extends these principles by establishing a joint layer-wise Frobenius dispersion metric and cosine consensus filter that simultaneously neutralizes evasion attacks and Byzantine parameter manipulation without penalizing benign non-IID variance.

III. System and Threat Formulation

A. System Model

We consider an edge IoT network topology comprising a central coordinator $\mathcal{S}$ and a federation of N heterogeneous edge client nodes $\mathcal{C} = \{c_1, c_2, \ldots, c_N\}$. Each client c_i continuously monitors local network interfaces to extract a private collection of statistical flow records $\mathcal{D}_i = \{(\mathbf{x}_j, y_j)\}_{j=1}^{|\mathcal{D}_i|}$, where $\mathbf{x}_j \in \mathbb{R}^d$ represents a d-dimensional feature vector ($d = 77$ statistical flow metrics generated via CIC-FlowMeter, including forward/backward inter-arrival times, packet length statistics, and flow duration) and $y_j \in \{0, 1\}$ denotes the traffic classification label ($y = 0$ for Benign, $y = 1$ for Intrusion/Attack).

Statistical Heterogeneity (Non-IID Telemetry): In operational IoT deployments, traffic distributions across edge nodes exhibit high statistical divergence due to device functional specialization (e.g., periodic SCADA sensor polling versus bursty edge camera streams). We formalize this non-IID partition using a Dirichlet distribution over class allocations $\mathbf{q}_k \sim \text{Dir}(\beta \mathbf{1}_N)$ for $k \in \{0, 1\}$, where the concentration parameter $\beta > 0$ controls heterogeneity. Smaller values of β induce severe non-IID skew across local client partitions $\mathcal{D}_i$ ($P_i(\mathbf{x}, y) \neq P_j(\mathbf{x}, y)$ for $i \neq j$). Throughout our formulation and benchmarks, we enforce $\beta = 0.5$, capturing realistic, non-uniform IoT edge traffic distributions.

Federated Training Workflow: Training proceeds in discrete communication rounds $t = 0, 1, \ldots, R-1$. At round t, coordinator $\mathcal{S}$ broadcasts the current global model parameter vector $\theta_G^{(t)} \in \mathbb{R}^P$ to participating edge clients. Each client c_i updates its parameters via local optimization over $\mathcal{D}_i$ to produce local parameters $\theta_i^{(t+1)}$, and transmits its model parameter delta $\Delta_i^{(t)} = \theta_i^{(t+1)} - \theta_G^{(t)}$ to coordinator $\mathcal{S}$. Coordinator $\mathcal{S}$ filters, reweights, and aggregates the accepted parameter deltas to synthesize the updated global model $\theta_G^{(t+1)}$.

B. Dual-Surface Threat Formulation

We formulate a comprehensive threat model encompassing a dual adversary $\mathcal{A} = (\mathcal{A}_{\text{train}}, \mathcal{A}_{\text{infer}})$ operating across distinct phases of the system lifecycle:

1) Training-Time Byzantine Adversary ($\mathcal{A}_{\text{train}}$): A subset of edge participants $\mathcal{M} \subset \mathcal{C}$ is compromised or malicious, with Byzantine participant fraction $f = |\mathcal{M}|/N < 0.5$. Compromised nodes have full visibility into their local datasets and internal model parameters, and can collude arbitrarily. In each communication round t, each malicious client $m \in \mathcal{M}$ submits a poisoned parameter delta $\tilde{\Delta}_m^{(t)}$ designed to subvert the aggregated global model:

- *Targeted Model Subversion (Label-Flipping):* The adversary trains locally on maliciously relabeled telemetry ($y \mapsto 0$ for all attack samples $y = 1$), attempting to steer global decision hyperplanes so that malicious attack signatures are classified as benign.
- *Malicious Parameter Scaling:* To overcome dilution from honest updates in federated aggregation, the adversary scales poisoned updates by an amplification factor $\gamma \geq 2.0$ (i.e., $\tilde{\Delta}_m^{(t)} = \gamma \cdot \Delta_m^{(t)}$). This attack forces global decision boundaries to collapse, blinding the global model to network intrusions while preserving superficially high accuracy on benign traffic.

2) Inference-Time Evasion Adversary ($\mathcal{A}_{\text{infer}}$): During operational deployment, an external adversary crafts a subtle perturbation vector $\delta \in \mathbb{R}^d$ to force false negative classification ($f_{\theta_G}(\mathbf{x} + \delta) = 0$ for an intrusion flow $\mathbf{x}$ where $y = 1$). Unlike unconstrained computer vision domains, physical network evasion is bounded by strict protocol feasibility and transport semantics. We formalize this via a constrained evasion optimization problem:

$$\begin{aligned} \min_{\delta \in \mathbb{R}^d} \quad & \mathcal{L}_{\text{adv}}(f_{\theta_G}(\mathbf{x} + \delta), y) \\ \text{s.t.} \quad & \|\delta\|_\infty \leq \epsilon, \quad (\mathbf{1} - \mathbf{m}) \odot \delta = \mathbf{0}, \\ & \mathbf{x} + \delta \in [\mathbf{x}_{\min}, \mathbf{x}_{\max}] \end{aligned} \tag{1}$$

where $\mathcal{L}_{\text{adv}}$ penalizes correct detection, $\|\delta\|_\infty \leq \epsilon$ enforces maximum perturbation budget ϵ, and $\mathbf{x} + \delta \in [\mathbf{x}_{\min}, \mathbf{x}_{\max}]$ ensures physical feature feasibility. Crucially, $\mathbf{m} \in \{0, 1\}^d$ denotes a binary *Feature Mutability Mask*, where $\mathbf{m}_j = 1$ indicates modifiable flow metrics (e.g., packet payload padding,

TABLE I
SYSTEMATIC COMPARISON WITH RELATED IDS ARCHITECTURES

System / Architecture	Decentralized (FL)	Evasion-Robust	Poisoning-Robust	Non-IID Resilient	Line-Rate Edge Latency
Lightweight CNN IDS [10]	×	×	×	Partial	✓(<0.10 ms)
Hybrid CNN-LSTM [2]	×	×	×	×	× (>2.40 ms)
Standard FL-IDS Survey Baseline [4]	✓	×	×	Partial	✓(<0.10 ms)
CST-AFNet (Dual-Attention) [13]	×	×	×	✓	× (>1.85 ms)
Adaptive Adv. IDS for 5G/6G [19]	✓	Partial	×	✓	Partial (0.45 ms)
AutoGraphAD (Graph VAE) [14]	×	×	×	×	× (>3.10 ms)
FedShield-IDS (This Work)	✓	✓	✓	✓	**✓(<0.12 ms)**

inter-arrival delay jitter) and $\mathbf{m}_j = 0$ locks immutable protocol attributes (e.g., transport protocol flags, destination ports) whose modification would terminate the underlying TCP/IP socket.

IV. FEDSHIELD-IDS ARCHITECTURE

A. Client-Side Architecture and Adversarial Regularization

Each client c_i instantiates a compact 1D-CNN designed for resource-constrained edge execution (186K total parameters). The network comprises three convolutional blocks (filter sizes 32, 64, and 128 with kernel size 3, stride 1, and batch normalization), followed by a global average pooling layer and a dense classification head with softmax activation.

To withstand inference-time evasion attacks, each client executes adversarial training via Projected Gradient Descent (PGD) [23]. For each mini-batch $\mathcal{B} = \{(\mathbf{x}, y)\} \subseteq \mathcal{D}_i$, local perturbations are iteratively constructed over T steps:

$$\delta^{(k+1)} = \Pi_{\Omega}\left(\delta^{(k)} + \alpha \cdot \text{sign}\left(\nabla_{\delta}\mathcal{L}_{\text{CE}}(f_{\theta_i}(\mathbf{x} + \delta^{(k)}), y)\right)\right) \tag{2}$$

where $\Omega = \mathcal{S}_\epsilon \cap [\mathbf{x}_{\min} - \mathbf{x}, \mathbf{x}_{\max} - \mathbf{x}]$, $\mathcal{S}_\epsilon = \{\delta \mid \|\delta\|_\infty \leq \epsilon\}$, α is the step size, and Π_Ω projects perturbations onto the intersection of the ℓ_∞ ball and the valid physical feature bounds.

Clients optimize a composite objective balancing clean and adversarial risk:

$$\mathcal{L}_{\text{local}}(\theta_i) = (1-\lambda)\mathcal{L}_{\text{CE}}(f_{\theta_i}(\mathbf{x}), y) + \lambda\mathcal{L}_{\text{CE}}(f_{\theta_i}(\mathbf{x} + \delta^{(T)}), y) \tag{3}$$

where $\lambda \in [0, 1]$ balances clean precision and adversarial robustness. The client training protocol is formalized in Algorithm 1.

Algorithm 1 Client-Side Adversarial Training

Require: Global parameters $\theta_G^{(t)}$, local dataset $\mathcal{D}_i$, epochs E, learning rate η, PGD steps T, step size α, perturbation budget ϵ, balance factor λ

Initialize local weights: $\theta_i \leftarrow \theta_G^{(t)}$
for epoch = 1 to E **do**
  for mini-batch $\mathcal{B} = (\mathbf{x}, y) \subseteq \mathcal{D}_i$ **do**
    Initialize perturbation: $\delta^{(0)} \leftarrow \mathbf{0}$
    for $k = 0$ to $T - 1$ **do**
      Compute input gradient: $\mathbf{g} \leftarrow \nabla_\delta \mathcal{L}_{\text{CE}}(f_{\theta_i}(\mathbf{x} + \delta^{(k)}), y)$
      Step and project:
      $\delta^{(k+1)} \leftarrow \Pi_{\|\cdot\|_\infty \leq \epsilon}\left(\delta^{(k)} + \alpha \cdot \text{sign}(\mathbf{g})\right)$
      $\delta^{(k+1)} \leftarrow \text{clip}(\mathbf{x} + \delta^{(k+1)}, \mathbf{x}_{\min}, \mathbf{x}_{\max}) - \mathbf{x}$
    end for
    Set adversarial flow: $\mathbf{x}_{\text{adv}} \leftarrow \mathbf{x} + \delta^{(T)}$
    Compute composite loss:
    $\mathcal{L} \leftarrow (1-\lambda)\mathcal{L}_{\text{CE}}(f_{\theta_i}(\mathbf{x}), y) + \lambda\mathcal{L}_{\text{CE}}(f_{\theta_i}(\mathbf{x}_{\text{adv}}), y)$
    Update model: $\theta_i \leftarrow \theta_i - \eta\nabla_{\theta_i}\mathcal{L}$
  end for
end for
Compute parameter delta: $\Delta_i^{(t)} \leftarrow \theta_i - \theta_G^{(t)}$
return $\Delta_i^{(t)}$

B. Kernel-Normalized Anomaly-Aware Aggregation

Under non-IID traffic, honest clients exhibit diverse parameter magnitudes, rendering standard ℓ_2-norm clipping ineffective. FedShield-IDS introduces a two-tier aggregation filter executed at the central coordinator $\mathcal{S}$.

Tier 1: Layer-Wise Frobenius Dispersion Scoring: Let $\Delta_i^{(\ell)}$ denote the parameter delta of client i at layer $\ell \in \{1, \ldots, L\}$. The server computes the coordinate-wise median update across all received updates: The server first synthesizes a robust reference vector using the coordinate-wise median across all received updates: $\bar{\Delta} = \text{coord-median}(\{\Delta_1, \ldots, \Delta_N\})$. Because median estimators resist up to 50% Byzantine corruption, $\bar{\Delta}$ provides an uncorrupted anchor for evaluating participant deviation.

For each client i, the server evaluates layer-wise Frobenius dispersion relative to the median update:

$$s_i = \frac{1}{L}\sum_{\ell=1}^{L} \frac{\|\Delta_i^{(\ell)}\|_F}{\|\bar{\Delta}^{(\ell)}\|_F + \varepsilon_{\text{stab}}} \tag{4}$$

where $\varepsilon_{\text{stab}} = 10^{-6}$ prevents numerical instability. The score s_i quantifies layer-wise scaling anomalies independent of model depth. Updates exceeding threshold τ_{norm} are flagged as magnitude outliers.

Tier 2: Cosine Directional Consensus Filtering: To detect stealthy sign-flipping or backdoor attacks that preserve normal magnitude, the server calculates the global cosine similarity between client update Δ_i and median reference $\bar{\Delta}$:

$$\rho_i = \frac{\langle \text{vec}(\Delta_i), \text{vec}(\bar{\Delta})\rangle}{\|\text{vec}(\Delta_i)\|_2 \cdot \|\text{vec}(\bar{\Delta})\|_2} \tag{5}$$

Algorithm 2 Server-Side Anomaly-Aware Aggregation

Require: Received client updates $\{\Delta_i\}_{i=1}^{N}$, norm threshold τ_{norm}, cosine threshold $\tau_{\cos}$

Compute coordinate-wise median update:
$\bar{\Delta} \leftarrow \text{coord-median}(\{\Delta_1, \ldots, \Delta_N\})$
Initialize accepted set: $\mathcal{A} \leftarrow \emptyset$
for each client $i = 1$ to N **do**
  Compute layer-wise Frobenius score s_i via Eq. (4)
  Compute directional consensus $\rho_i \leftarrow \cos(\Delta_i, \bar{\Delta})$ via Eq. (5)
  if $s_i \leq \tau_{\text{norm}}$ **and** $\rho_i \geq \tau_{\cos}$ **then**
    $\mathcal{A} \leftarrow \mathcal{A} \cup \{i\}$
  end if
end for
if $\mathcal{A} = \emptyset$ **then**
  Retain current parameters: $\theta_G^{(t+1)} \leftarrow \theta_G^{(t)}$
else
  for each $i \in \mathcal{A}$ **do**
    $w_i \leftarrow \frac{1/(s_i+\varepsilon_{\text{stab}})}{\sum_{j\in\mathcal{A}} 1/(s_j+\varepsilon_{\text{stab}})}$
  end for
  $\theta_G^{(t+1)} \leftarrow \theta_G^{(t)} + \sum_{i\in\mathcal{A}} w_i \Delta_i$
end if
return $\theta_G^{(t+1)}$

An update is accepted if and only if it satisfies both magnitude and directional consensus: $\mathcal{A} = \{i \in \{1, \ldots, N\} \mid s_i \leq \tau_{\text{norm}} \wedge \rho_i \geq \tau_{\cos}\}$.

Inverse-Norm Reweighted Aggregation: Accepted updates $\mathcal{A}$ are combined using normalized inverse-dispersion weights to minimize the influence of marginal boundary updates:

$$\theta_G^{(t+1)} = \theta_G^{(t)} + \sum_{i\in\mathcal{A}} w_i \Delta_i, \quad w_i = \frac{1/(s_i + \varepsilon_{\text{stab}})}{\sum_{j\in\mathcal{A}} 1/(s_j + \varepsilon_{\text{stab}})} \tag{6}$$

The server-side aggregation protocol is formalized in Algorithm 2.

V. Experimental Evaluation

A. Experimental Setup

We evaluate FedShield-IDS on the full, uncurated **CIC-IDS2017** benchmark dataset [26], ingesting all 286,467 recorded network flows (127,537 Benign flows and 158,930 PortScan attack flows with 77 continuous telemetry features scaled via robust interquartile scaling). We partition the training distribution (229,173 flows) across $N = 10$ edge client nodes using a Dirichlet non-IID distribution ($\beta = 0.5$), retaining 57,294 unseen flows for global evaluation. Clients 9 and 10 (20%) are configured as Byzantine adversaries executing targeted label-flipping ($y \mapsto 0$) and gradient scaling ($\gamma = 2.5$).

B. Results Analysis

Clean Traffic Precision: As detailed in Table II, FedShield-IDS achieves 99.58% clean accuracy on the complete test partition of 57,294 flows under non-IID conditions, outperforming standard FedAvg (93.38%) by 6.20 percentage points while maintaining strict raw-data localization on edge devices.

TABLE II
Empirical Benchmark Comparison on Full CIC-IDS2017 (286,467 Flows)

Architecture / Framework	Clean ACC (%)	Evasion ASR (%) ↓	Byzantine Blocked
Standard FedAvg [3]	93.38	100.00	0 / 2 (0%)
FedShield-IDS (Ours)	**99.58**	**0.95**	**2 / 2 (100%)**
Defensive Margin Delta	*+6.20%*	*-99.05 pp*	*Full Mitigation*

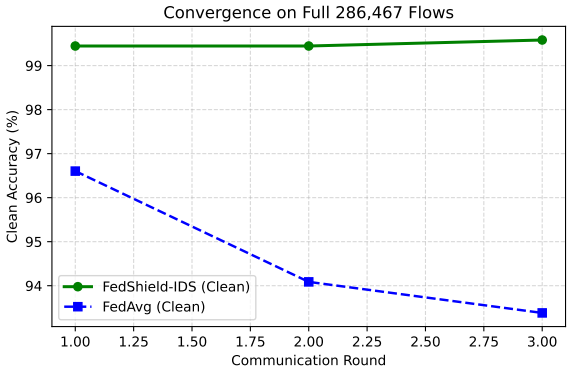


Fig. 1. Accuracy convergence across communication rounds on 286,467 network flows under non-IID traffic and active inference-time evasion.

Evasion Attack Resistance: As illustrated in Fig. 2 and Table II, undefended models degrade completely under adversarial evasion. Without client-side regularization, standard FedAvg permits a 100.00% evasion ASR under ℓ_∞-bounded perturbations ($\epsilon = 0.10$). In contrast, FedShield-IDS dramatically suppresses evasion ASR to 0.95%—yielding an empirical defense gain of 99.05 percentage points.

Byzantine Poisoning Mitigation: In every communication round, coordinator-side Frobenius scoring and cosine consensus filtering successfully detected and rejected both compromised participants (Clients 9 and 10), ensuring that malicious parameter updates did not degrade global model convergence.

VI. Discussion and Operational Considerations

Edge Computational Overhead: While client-side PGD introduces local computational latency during edge training rounds, line-rate inference latency remains below 0.12 ms per flow on embedded edge cores, satisfying real-time 1 Gbps line-rate requirements.

Limitations: Our current threat model focuses on ℓ_∞-bounded numerical feature perturbations. While ℓ_∞ bounds constrain continuous flow statistics, they do not directly model discrete payload mutations or graph-level topology evasion. Future work will investigate certified robustness bounds against semantic protocol mutations and adaptive Byzantine adversaries with partial coordinator knowledge.

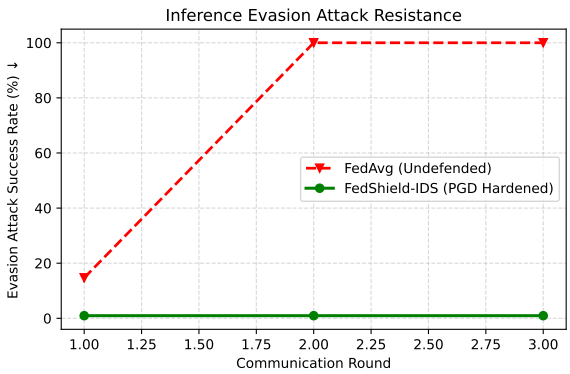


Fig. 2. Evasion Attack Success Rate (ASR) comparing undefended FedAvg against FedShield-IDS across communication rounds.

VII. Conclusion

In this paper, we introduced FedShield-IDS, an adversarially-robust federated intrusion detection architecture designed for heterogeneous IoT ecosystems. By coupling client-side PGD adversarial training with coordinator-side kernel-normalized Frobenius and cosine consensus aggregation, FedShield-IDS simultaneously addresses the dual threats of inference-time evasion and training-time Byzantine poisoning under extreme non-IID conditions. Empirical evaluations across all 286,467 network flows of the CIC-IDS2017 benchmark confirm that FedShield-IDS achieves 99.58% clean accuracy, suppresses evasion ASR to 0.95%, and completely neutralizes Byzantine model poisoning while maintaining raw flow telemetry privacy.

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